

Automotive Intelligence – Closing the Information Gap via telemetry



By Ayush Agarwal

Disclaimer - The author does not represent the mentioned companies, the views presented are personal

<https://www.linkedin.com/in/ayush-agarwal-261041215/>

The Automotive Information Gap

Key Insight - Drivers already buy tools to understand their cars, , but current solutions are fragmented

Fuel Costs Are Poorly Tracked



Most drivers **estimate fuel spending** instead of tracking it accurately.

Common workarounds include:

- Rough mileage tracking
- Dedicated fuel payment cards

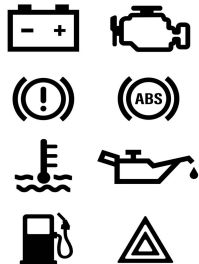
Manufacturer App Discovery Problem



Even when official apps exist:

- Many drivers **never install them**
- Some are **not aware they exist**

Vehicle Health Gap

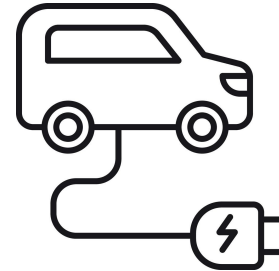


Vehicle Health Visibility Is Weak

Drivers receive alerts **too late**.

“Check Engine” lights only appear **after problems occur**, leaving drivers dependent on mechanics for diagnosis.

Third-Party Add-Ons (GPS + Dashcam)



Drivers often install external tools like **dashcams and GPS trackers** to gain visibility into their vehicles.

Signal: strong demand for vehicle intelligence.

Solving the Discovery Problem

TaaS – Telemetry as a Service - Getting Drivers Into the Ecosystem

Step 1 - Scan -

User scans QR code on dashboard.

Opens lightweight webapp dashboard instantly (No app install friction.)

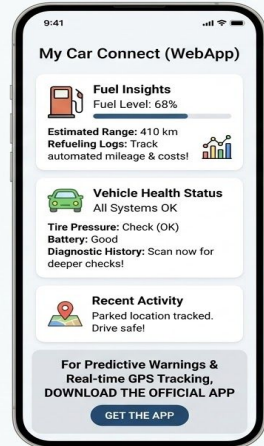


Step 2 — Immediate Value

Web dashboard shows:

- Fuel usage insights
- Vehicle health status
- Location information

User immediately sees **value**.

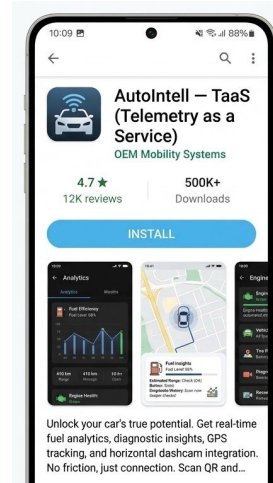


Wait my car can do this?



Step 3 — App Conversion

Prompt appears: “Enable real-time alerts and deeper insights — download the official car app.”



Step 4 — Real-Time Vehicle Alerts

Users receive proactive alerts about their vehicle
Examples:

- Low fuel risk
- Possible engine issues
- Security alerts



Fuel Cost Intelligence

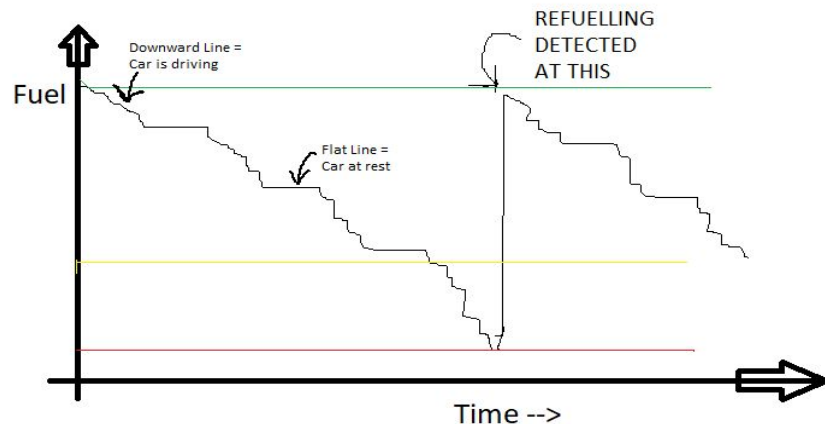
OBD - On Board Diagnostics

Modern cars expose real-time telemetry through the OBD interface. Key signals we need -

- Fuel level
- Odometer readings



Detecting Fuel Consumption -



Converting Fuel Data Into Money Insights -

Fuel Spending = Fuel Used (Liters) × Local Fuel Price

$$\text{Avg. Mileage} = \frac{\text{Distance Travelled (km)}}{\text{Fuel used (L)}}$$

Monthly Analytics -

Users receive **monthly fuel reports** to track driving cost trends.

Monthly Fuel Summary

Distance: 1,402 km

Fuel Used: 92 L

Fuel Cost: ₹9,384



Beyond the Check Engine Light: Predictive Engine Health

From reactive repairs to proactive vehicle care.

User Needs -

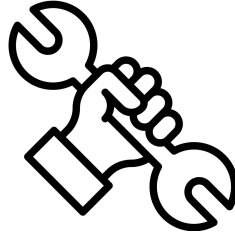
Predict failures early



Diagnose what broke

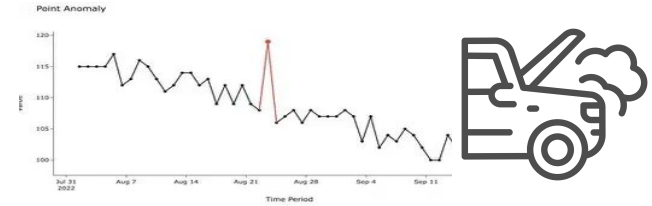


Find nearest repair center

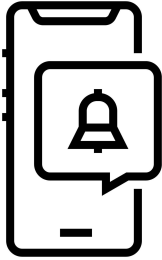


Technical Feasibility -

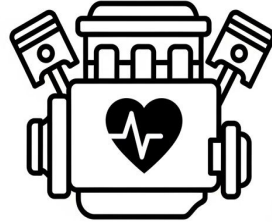
Using AI and IoT on Engine parameters obtained from OBD Data, we can get insights to solve these problems



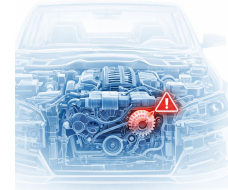
Solutions -



Notifications can be sent via App to inform users about **potential future breakdowns**



Monthly **engine health reports** can be shared via **email**.



Engine Diagnostic
! Component Failure Detected
Fault:
Alternator
Issue:
Alternator not charging battery
Recommended Action
Replace alternator

App can also display the **failed component + nearest service center**

Where Is My Car? Remote Vehicle Security

Peace of mind through location tracking and incident recording

Why Built-In OEM Integration Matters -

Seamless setup – devices pre-integrated with car systems

Trusted hardware – sourced by manufacturer

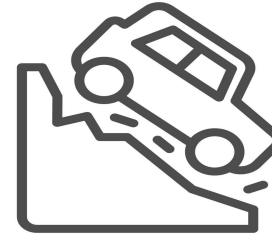
Unified app experience – no third-party apps needed

User Needs -

Where is my car parked?



Who damaged my car?

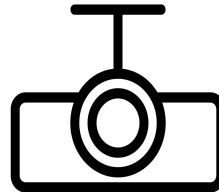


Security Features -



GPS Tracking -

- Real-time vehicle location
- Track parked vehicle
- Recover stolen vehicles



Dashcam Protection -

- Record incidents automatically
- Evidence during disputes
- Cloud storage for footage

Appendix